

Predictive-Coding-Inspired Reconstruction of Nonlinear Dynamics in Spiking Neural Networks

Bulat B. Batuev ^{1,*} , Sergey V. Sukhov ² 

¹ Kotelnikov Institute of Radioengineering and Electronics of the Russian Academy of Sciences, Moscow 125009, Russia; batuyev_bb@cplire.ru

² Ulyanovsk Branch, Kotelnikov Institute of Radio Engineering and Electronics of the Russian Academy of Sciences, Ulyanovsk 432071, Russia; ssukhov@ulireran.ru

* Correspondence: batuyev_bb@cplire.ru

Abstract

Accurate reconstruction of nonlinear time-dependent dynamics is a fundamental challenge for recurrent spiking neural networks and neuromorphic signal-processing systems. In this work, we compare two prediction-error architectures for nonlinear state reconstruction: a model with an external comparator and a model employing a dedicated spiking population to encode prediction error. Both architectures are implemented within the Neural Engineering Framework and trained using the local Prescribed Error Sensitivity learning rule. The proposed approach is evaluated on a two-dimensional periodic oscillator and the nonlinear Lorenz system. The reconstruction fidelity is assessed using the reconstruction error together with amplitude-envelope correlation, instantaneous-frequency correlation, phase-locking value, mean phase difference, and computational efficiency. The results demonstrate that both architectures accurately reconstruct nonlinear dynamics, while the spiking error representation provides a larger stable operating region and a more explicit spike-based representation of the bottom-up prediction-error pathway.

Keywords: predictive coding; spiking neural network; neural engineering framework; local learning; Lorenz attractor; state reconstruction; nonlinear dynamics

1. Introduction

The synchronization phenomena in natural and engineering systems attracted continuous attention over several decades. Spiking neural networks (SNNs) provide an event-based framework for representing and processing continuous time-dependent signals. A central technical challenge for SNNs synchronization is the reproduction of autonomous nonlinear dynamics and the reconstruction of another population state using local learning rules. One possible solution to this problem may be achieved with prediction-error-based hierarchical architectures. A higher-level population in these architectures generates predictions, whereas a lower-level population is the one whose activity is predicted and where the corresponding prediction error is computed [1,2]. The idea of these architectures is inspired by a predictive coding (PC) concept. The essence of PC is in the minimization of the prediction error between the predicted and actual signals coming into one of the layers of a neural network. The classic architecture of a neural network with PC uses separate neural populations to calculate the mismatch between the activity at the lower levels of the neural network and the predictions generated at a higher level of the hierarchy [1]. At the same time, the higher levels in the neural hierarchy send their predictions via feedback

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connections to the lower levels. Only the prediction error is sent up the hierarchy from the lower levels.

Here we develop SNN models in which prediction-error feedback is used to reconstruct an autonomous low-dimensional state generated by another spiking population. To model SNN dynamics, we use the Neural Engineering Framework, which allows us to build and simulate large-scale models suitable for both robotic applications [3] and for understanding and explaining neurobiological experiments [4].

We compare two implementations of the prediction-error architecture. In the external-comparator variant, the mismatch between the autonomous state and its reconstruction is obtained by direct subtraction outside the spiking substrate and supplied to the latent-state dynamics as a continuous signal. In the spiking-comparator variant, the same mismatch is encoded by a dedicated neuronal population, and its decoded activity drives latent dynamics.

To bring the reconstruction in line with the actual signal, the corresponding synaptic connections need to be modified (the network needs to be trained). In spiking neural networks, training is done locally at the level of individual synapses. This distinguishes SNNs from artificial neural networks and presents an additional challenge: methods for building and training SNNs are still actively studied, particularly regarding the locality and stability of local synaptic learning rules [5–7].

The central question addressed in this work is how the implementation of the prediction-error pathway affects the reconstruction of autonomously generated nonlinear dynamics in a spiking neural network. To this end, we (i) formulate two spiking predictive-reconstruction architectures that differ in how the prediction error is represented in the bottom-up pathway; (ii) quantify the fidelity with which the top-down prediction reconstructs the autonomous sensory dynamics, using amplitude, instantaneous-frequency, and phase measures. From a dynamical-systems standpoint, the object of study is a coupled system – an autonomous sensory oscillator (or a chaotic Lorenz attractor) coupled to a higher-level predictor through the predictive-coding loop – so that the reconstruction is produced by the interaction of two neural populations rather than imposed by external forcing. These two benchmark systems were selected to represent qualitatively different dynamical regimes. The two-dimensional oscillator provides a simple periodic system for which amplitude, frequency, and phase relationships can be interpreted directly. The Lorenz system provides a canonical nonlinear chaotic system with sensitivity to initial conditions and a substantially more complex phase-space structure. The purpose of using these two systems was to determine whether the same predictive-reconstruction architecture can operate in both regular and chaotic regimes.

The proposed error-mediated spiking reconstruction may be useful in neuromorphic control systems, adaptive sensorimotor systems, and event-based signal processors. Spike-based predictive reconstruction provides a mechanism for low-power reconstruction and tracking of dynamical signals on neuromorphic hardware [1,3]. An explicit spiking error pathway also provides a basis for future investigation of neuronal noise, spike loss, sparse communication, and heterogeneous neural parameters.

2. Related work

A significant number of studies have been devoted to the collective dynamics of recurrent networks and to the coordinated firing they can produce. For example, in [8], the classical model of sparse mixed excitatory-inhibitory networks showed that even simple static networks can generate a wide variety of regimes, from asynchronous irregular activity to coherent population oscillations. Such spike-level coherence should be distinguished from the problem considered here, which is posed at the level of the decoded output.

The quantity of interest in the present work is the low-dimensional signal decoded from one neural population and the accuracy with which a second population reproduces it, rather than the correlation structure of the spike trains themselves. Reconstruction in this output-level sense is the formulation commonly adopted in neuroengineering problems dealing with chaotic systems.

The development of theoretical ideas about PC in recent years has led to the formation of a unified framework linking Bayesian inference, hierarchical neural models, and numerous experimental data. The review [2] systematizes various formulations of predictive coding, discussing both continuous and discrete models, the connection with variational inference, and the free energy principle. Based on these theoretical ideas, several models have been proposed in recent years that combine predictive coding with artificial neural networks. The paper [9] presented a model that combines predictive coding and reservoir computing in cultured neurons to process sensory information. The PC-SNN model proposed in [6] applies predictive coding to supervised learning tasks in SNNs. The research presented in [5] considers a spiking version of predictive coding for continuous learning on data streams, emphasizing the importance of learning rule locality and stability during long-term sequential learning. The work [10] applied predictive coding to unsupervised learning of MNIST digit classification. Finally, the research in [7] showed that SNNs with a random structure spontaneously learn to predict temporal sequences of stimuli based solely on the adjustment of synaptic weights using the local STDP (spike-timing-dependent plasticity) method.

In parallel, a direction related to neuroengineering approaches to the construction and analysis of recurrent neural networks was developed. Ref. [11] investigated the fine-tuning and stability properties of PES (Prescribed Error Sensitivity) learning rule for recurrent networks, showing that small changes in parameters can lead to significant shifts in dynamics, and that the correct procedure for selecting decoders and recurrent weights is critical for maintaining stability. The works [12] and [13] offer a detailed technical overview of the Neural Engineering Framework (NEF) and demonstrate its application, including to the modeling of action selection in the basal ganglia.

Several approaches have used error feedback to train recurrent spiking networks to reproduce nonlinear dynamics. In [14], simple spiking architectures containing predictive-coding elements – integrators, driven oscillators, and coupled oscillators built from populations of leaky integrate-and-fire (LIF) neurons – were proposed for hierarchical control of dynamical systems. Boerlin, Machens, and Denève [15] derived SNNs, in which neuronal membrane potentials represent projections of the population-level prediction error and spikes are emitted when they reduce the representational loss; their construction provides an efficient spike-based implementation of linear dynamical systems. Gilra and Gerstner [16] introduced the FOLLOW rule, in which output errors are projected back to heterogeneous spiking neurons through fixed connections and drive local online adaptation of feedforward and recurrent synapses, allowing nonlinear and chaotic dynamics to be learned and reproduced after removal of the error feedback. Nicola and Clopath [17] developed a supervised learning method of recurrent dynamics (FORCE) that trains spiking recurrent networks to generate target dynamics, including complex dynamical trajectories. Bourdoukan and Denève [18] demonstrated that balanced recurrent spiking networks can learn linear dynamics using local plasticity rules, with excitation–inhibition balance making a global error signal locally available to individual neurons. Alemi et al. [19] subsequently extended this framework to arbitrary nonlinear dynamics by combining efficient balanced networks with adaptive-control principles, obtaining local error-driven learning together with sparse activity and robustness to perturbations. These studies establish that local error

feedback can be used to train recurrent spiking networks to reproduce linear and nonlinear dynamics.

In comparison to previous works, we address a specific architectural question: how the implementation of the prediction-error comparator affects reconstruction dynamics. We therefore compare two otherwise equivalent NEF-based architectures in which the same prediction error is either calculated externally as a continuous signal or represented by a dedicated spiking population, while keeping the remaining network structure and learning mechanism unchanged. This controlled comparison makes it possible to isolate the effect of spike-mediated error representation on reconstruction fidelity and on the stable operating range of the predictive-error loop.

3. Methods

3.1. Modeling neural dynamics

To model dynamics in spiking neural networks, we used the Neural Engineering Framework (NEF) [4]. NEF is based on three principles that allow for the simple construction of large-scale neural models. According to the first principle, a population of neurons with activity a_i (index i numbers the neurons) collectively represents the value of a time-varying vector u . The relationship between the representation vector u and the underlying neural activity a_i is defined by

$$a_i = G(k_i \langle e_i, u \rangle + b_i), \quad (1)$$

where $G(\cdot)$ is a nonlinear function characteristic of the neuron used for modeling, k_i are the amplification parameters, $\langle e_i, u \rangle$ denotes the inner (dot) product of vectors e_i and u , e_i are the encoding vectors, and b_i are the constant background bias currents. The estimate of the vector u can be found from the activities a_i as

$$\hat{u} = \sum_i a_i d_i, \quad (2)$$

where d_i are linear decoders. Thanks to the second principle, NEF allows one to quickly compute linear and nonlinear functions $f(u)$ on represented vectors. Finally, the principle of dynamic behavior construction enables neural ensembles to simulate dynamics using recurrent connections [11]:

$$\frac{du}{dt} = Au + B\gamma. \quad (3)$$

Here, A and B are matrices that determine the required dynamics. Stimuli γ are the external control signals that are fed into the network through matrix B . To properly model equation (3) using ensembles of neurons in accordance with NEF, the weights of recurrent connections must compute the function $\tau Au + u$, and the weights of input connections must compute the function $\tau B\gamma$, where τ is the synaptic time constant [12].

3.2. Predictive coding inspired neural network model

In this work, two related predictive-coding-inspired architectures were compared (Figure 1). The first architecture (Figure 1(a)), denoted as a predictive-coding architecture with an external comparator (PC-EC), was used as a baseline in which the prediction error was computed as a continuous vector and directly injected into the latent dynamics. The second architecture (Figure 1(b)), denoted as predictive-coding architecture with a spiking comparator (PC-SC), represents the prediction error in a separate spiking population.

Let $u_{\text{ref}}(t) \in \mathbb{R}^D$ denote the reference input signal, where t is continuous time and D is the dimension of the represented dynamical system. In the experiments, $D = 2$ for the oscillator and $D = 3$ for the Lorenz attractor. The sensory population o_1 first

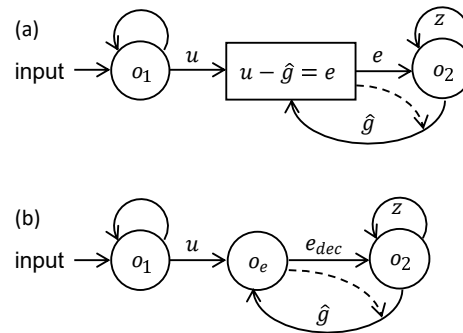


Figure 1. Two implementations of the proposed predictive-error architecture. (a) In the external-comparator variant (PC-EC), the mismatch $e = u_{o_1} - \hat{g}$ is computed by an algebraic subtraction block and supplied directly to the latent-state dynamics. (b) In the spiking-comparator variant (PC-SC), the sensory state and top-down reconstruction are projected with opposite signs onto a dedicated error population, whose decoded activity e_{dec} provides a spike-mediated estimate of the same mismatch. The dashed arrow indicates the applied local learning.

received the reference signal and was equipped with an autonomous recurrent decoder. The corresponding sensory state of the population o_1 is denoted as u_{o_1} . The higher level population o_2 has its own dynamics with latent state $z[n]$ and the raw reconstructed state $\hat{g}$. Through the feedback connection, the reconstructed state $\hat{g}$ is compared to u_{o_1} , and the disagreement $e[n]$ is calculated either by external comparator (Figure 1(a)), or by separate neural population o_e (Figure 1(b)). The prediction error $e[n]$ is then supplied forward to modify the dynamics of o_2 and to improve the prediction. The main distinction between PC-EC and PC-SC is therefore the representation of the prediction error in the bottom-up pathway. In PC-EC, the latent state receives the continuous error vector $e[n]$ directly (Figure 1(a)). In PC-SC, this error is encoded by a separate spiking population and the latent state receives the decoded estimate $e_{dec}[n]$ (Figure 1(b)). Consequently, PC-SC provides a more explicit spiking implementation of predictive coding, in which sensory representation, top-down reconstruction, and prediction-error representation are assigned to distinct neural populations.

The explicit error population in PC-SC may be interpreted as a functional analogue of prediction-error populations proposed in cortical predictive-processing models [20, 21]. Nevertheless, the population o_e used here should not be identified with a particular cortical layer or cell type. It represents the computational function of explicitly encoding a prediction error. Consequently, our results establish the computational consequences of a spike-mediated error representation rather than a particular cortical implementation.

For numerical simulations, time was discretized as $t_n = n\Delta t$, where n is the discrete time-step index and Δt is the simulation time step. Accordingly, $u_{ref}[n] \equiv u_{ref}(t_n)$.

The recurrent decoder was trained to approximate the NEF dynamical target

$$u_{target}[n] = u_{ref}[n] + \tau_{syn}\dot{u}_{ref}[n], \quad (4)$$

where $u_{target}[n] \in \mathbb{R}^D$ is the target vector for the recurrent decoder, τ_{syn} is the time constant of the autonomous sensory dynamics, and $\dot{u}_{ref}[n]$ is the discrete-time approximation of the time derivative of the reference signal. The derivative was approximated as

$$\dot{u}_{ref}[n] \approx \frac{u_{ref}[n] - u_{ref}[n-1]}{\Delta t}. \quad (5)$$

Thus, the recurrent decoder was obtained from the following regularized least-squares problem:

$$W_{\text{rec}} = \arg \min_W \sum_{n \in T_{\text{train}}} \left\| W^\top \alpha_{o_1}[n] - u_{\text{target}}[n] \right\|_2^2 + \lambda_{\text{rec}} \|W\|_F^2. \quad (6)$$

Here, T_{train} denotes the set of discrete time indices used for training the recurrent decoder, $W \in \mathbb{R}^{N_{\text{sens}} \times D}$ is the decoder matrix optimized in the least-squares problem, $W_{\text{rec}} \in \mathbb{R}^{N_{\text{sens}} \times D}$ is the trained recurrent decoder, $\alpha_{o_1}[n] \in \mathbb{R}^{N_{\text{sens}}}$ is the filtered spike activity of the sensory population, N_{sens} is the number of neurons in the sensory population, λ_{rec} is the ridge regularization parameter, and $\|\cdot\|_F$ denotes the Frobenius norm.

After initialization, the autonomous sensory state was updated as

$$u_{o_1}[n] = u_{o_1}[n-1] + \Delta t \dot{u}_{o_1}[n], \quad (7)$$

where $\dot{u}_{o_1}[n]$ is the time derivative of the autonomous sensory state, approximated in discrete time as

$$\dot{u}_{o_1}[n] = \frac{-u_{o_1}[n-1] + r_{\text{rec}}[n] + c[n]}{\tau_{\text{syn}}}. \quad (8)$$

Here,

$$r_{\text{rec}}[n] = W_{\text{rec}}^\top \alpha_{o_1}[n] \quad (9)$$

is the recurrent reconstruction decoded from the filtered activity of the sensory population, and $c[n] \in \mathbb{R}^D$ is the cue signal used during the initial forcing interval. The cue signal was defined as

$$c[n] = \begin{cases} u_{\text{ref}}[n-1] - u_{o_1}[n-1], & n\Delta t < T_{\text{cue}}, \\ \mathbf{0}, & n\Delta t \geq T_{\text{cue}}, \end{cases} \quad (10)$$

where T_{cue} is the duration of the initial cue-forcing interval and $\mathbf{0} \in \mathbb{R}^D$ is the zero vector. During the cue interval, u_{o_1} was therefore guided toward the reference trajectory. After T_{cue} , the cue was removed, and the sensory population evolved autonomously according to its learned recurrent dynamics. Consequently, in the autonomous regime, the reconstruction target for the higher-level population was not the externally supplied reference u_{ref} , but the internally generated sensory state u_{o_1} .

The top-down reconstruction produced by the higher-level population o_2 was decoded as

$$\hat{g}[n] = W_{o_2\text{pred}}^\top \alpha_{o_2}[n], \quad (11)$$

where $\alpha_{o_2}[n] \in \mathbb{R}^{N_{\text{lat}}}$ is the filtered spike activity of the latent population, N_{lat} is the number of neurons in the latent population, and $W_{o_2\text{pred}} \in \mathbb{R}^{N_{\text{lat}} \times D}$ is the trainable reconstruction decoder. To approximate postsynaptic integration, $\hat{g}[n]$ was additionally low-pass filtered:

$$g[n] = g[n-1] + \frac{\Delta t}{\tau_{\text{syn}}} (\hat{g}[n] - g[n-1]). \quad (12)$$

The prediction error $e[n] \in \mathbb{R}^D$ measures the mismatch between the autonomous sensory state and its top-down reconstruction:

$$e[n] = u_{o_1}[n] - \hat{g}[n]. \quad (13)$$

In the PC-EC architecture, this continuous error vector was directly used to drive the latent dynamics:

$$z[n] = z[n-1] + \Delta t \left(-\frac{z[n-1]}{\tau_{\text{syn}}} + e[n] \right), \quad (14)$$

where τ_{syn} is the latent-state time constant.

The latent state was then restricted to the representational range of the neural population by the component-wise projection

$$z[n] = \text{clip}(z[n], -1, 1), \quad (15)$$

where

$$\text{clip}(z_i, -1, 1) = \min\{1, \max\{-1, z_i\}\}. \quad (16)$$

Here, z_i denotes the i -th component of the latent state vector $z[n]$. Thus, in PC-EC, the bottom-up pathway does not include an additional spiking representation of the error; the latent state receives the continuous vector $e[n]$ directly.

In the PC-SC architecture, the filtered activity of a separate LIF neural population o_e was decoded into a spike-based estimate of the prediction error:

$$e_{\text{dec}}[n] = W_e^\top \alpha_e[n], \quad (17)$$

where $\alpha_e[n] \in \mathbb{R}^{N_{\text{err}}}$ is the filtered spike activity of the error population, N_{err} is the number of neurons in the error population, and $W_e \in \mathbb{R}^{N_{\text{err}} \times D}$ is its fixed decoder. The decoder W_e was obtained by a regularized least-squares fit on sampled error vectors $\xi_m \in [-1, 1]^D$:

$$W_e = \arg \min_W \sum_m \left\| W^\top \alpha_e(\xi_m) - \xi_m \right\|_2^2 + \lambda_e \|W\|_F^2. \quad (18)$$

Here, ξ_m is the m -th sampled error vector, $\alpha_e(\xi_m)$ denotes the filtered activity of the error population evoked by the input vector ξ_m , $W \in \mathbb{R}^{N_{\text{err}} \times D}$ is the optimized decoder matrix, and λ_e is the ridge regularization parameter for the error decoder.

The latent dynamics were then driven by the decoded spiking error estimate:

$$z[n] = z[n-1] + \Delta t \left(-\frac{z[n-1]}{\tau_{syn}} + e_{\text{dec}}[n] \right). \quad (19)$$

This update was followed by the same bounded-state projection, $z[n] = \text{clip}(z[n], -1, 1)$.

In both architectures, the reconstruction decoder of the higher-level population was adapted using a PES-like local learning rule [11]:

$$\Delta W_{o_2 \text{pred}}[n] = \eta \alpha_{o_2}[n] e^\top[n] \Delta t, \quad (20)$$

where $\Delta W_{o_2 \text{pred}}[n] \in \mathbb{R}^{N_{\text{lat}} \times D}$ is the decoder update at time step n , η is the learning rate, and $e^\top[n]$ is the transpose of the prediction-error vector. The decoder update is therefore proportional to the outer product between the filtered activity of the latent population and the continuous prediction error.

The reason that reducing the prediction error yields an accurate state reconstruction can be stated as follows. The local PES rule adapts the reconstruction decoder $W_{o_2 \text{pred}}$ in the direction that decreases the instantaneous error $e[n] = u_{o_1}[n] - \hat{g}[n]$, so that at a fixed point of this adaptation the top-down reconstruction $\hat{g}$ reproduces the sensory trajectory u_{o_1} on average. Because u_{o_1} is not an arbitrary target but the state of an autonomous oscillator (or, for the Lorenz system case, a chaotic attractor), reproducing this trajectory is equivalent to matching its amplitude, instantaneous frequency, and phase. Driving $e \rightarrow 0$ therefore does not merely reduce a pointwise distance: it forces the higher-level population to reproduce the temporal structure of the lower-level dynamics, which is precisely what the amplitude-, frequency-, and phase-based measures introduced below quantify. Moreover, in the autonomous regime the reconstruction target is the internally

generated sensory state u_{o_1} rather than the external reference u_{ref} , so that the signal being reconstructed is itself produced inside the network.

In the first set of experiments, the input signal is defined as a continuous two-dimensional process $u(t) = (x(t), y(t))$, where $u(t) \in \mathbb{R}^2$ is the oscillator state vector, and $x(t)$ and $y(t)$ are the coordinates of a point rotating uniformly in the plane.

The rotation of the phase vector over one time step is described by the matrix

$$M = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}, \quad \theta = \omega \Delta t. \quad (21)$$

Here, $M \in \mathbb{R}^{2 \times 2}$ is the discrete-time rotation matrix, θ is the rotation angle over one time step, and ω is the angular frequency of the oscillator. The oscillator trajectory was generated in discrete time according to

$$u[0] = (1, 0)^\top, \quad u[n] = Mu[n-1], \quad n = 1, \dots, N_T. \quad (22)$$

The choice of $\theta = \omega \Delta t$ ensures a periodic trajectory along the unit circle.

The second modeled dynamic system was the Lorenz system [22]. Its continuous dynamics are specified by the system of ordinary differential equations

$$\begin{aligned} \dot{x}(t) &= \sigma(y(t) - x(t)), \\ \dot{y}(t) &= \rho x(t) - x(t)z(t) - y(t), \\ \dot{z}(t) &= x(t)y(t) - \beta z(t), \end{aligned} \quad (23)$$

where $u(t) = (x(t), y(t), z(t))^\top \in \mathbb{R}^3$ is the state vector of the Lorenz system, and $x(t)$, $y(t)$, and $z(t)$ are its coordinates. The parameters σ , ρ , and β are the standard Lorenz system parameters and were chosen in the chaotic regime: $\sigma = 10$, $\rho = 28$, $\beta = 2.667$. The initial condition was $s[0] = (1, 1, 1)^\top$, where $s[n] = (x[n], y[n], z[n])^\top$ denotes the discrete-time Lorenz system state. Numerical integration was performed using the fourth-order Runge–Kutta method with time step Δt .

The obtained raw trajectory was denoted by

$$u_{\text{raw}}[n] = s[n]. \quad (24)$$

To match the input range to the neural population settings, component-wise maximum-absolute-value normalization was applied. For each coordinate $j = 1, \dots, D$, where $D = 3$ for the Lorenz system, the normalization scale factor was computed as

$$S_j = \max_{0 \leq n \leq N_T} |u_{\text{raw},j}[n]|, \quad (25)$$

The normalized trajectory was defined by

$$u_j[n] = \frac{u_{\text{raw},j}[n]}{S_j}, \quad j = 1, \dots, D. \quad (26)$$

Thus, the normalized input sequence was

$$u[n] = \left(\frac{u_{\text{raw},1}[n]}{S_1}, \frac{u_{\text{raw},2}[n]}{S_2}, \frac{u_{\text{raw},3}[n]}{S_3} \right)^\top. \quad (27)$$

3.3. Dynamics of neurons

To verify the dependence of reconstruction quality on the number of neurons, simulations of neural activity were performed for different sizes of the sensory population o_1 . For the PC-SC/PC-EC comparison, N_{sens} was varied from 10 to 560 in steps of 50, while the latent o_2 and error o_e populations were kept fixed. Encoders $e_i \in \mathbb{R}^D$ were generated from a normal distribution $\mathcal{N}(0, 1)$ followed by row normalization.

For each neuron, the intercept $x_{\text{int}}^{(i)} \in (-1, 1)$ and maximum firing rate $r_{\text{max}}^{(i)} \in [100, 200]$ Hz were sampled. The intercept $x_{\text{int}}^{(i)}$ is the value of the projected input $\langle e_i, \chi \rangle$ at which the neuron reaches the firing threshold, where $\chi \in \mathbb{R}^D$ denotes a generic represented input vector. Assuming the standard NEF threshold current $J_{\text{th}} = 1$, it is defined by

$$1 = \kappa_i x_{\text{int}}^{(i)} + \beta_i. \quad (28)$$

Here, κ_i and β_i are the gain and bias current parameters of neuron i , respectively. For vector-valued inputs, the relation

$$\langle e_i, \chi \rangle = x_{\text{int}}^{(i)} \quad (29)$$

defines a hypersurface orthogonal to e_i that separates inactive and active states of the neuron.

The corresponding input current is defined using the inverse LIF rate equation:

$$J(r) = \left(1 - \exp \left[-\frac{1/r - \tau_{\text{ref}}}{\tau_{\text{rc}}} \right] \right)^{-1}, \quad r > 0. \quad (30)$$

Here, $J(r)$ is the input current corresponding to the firing rate r , τ_{rc} is the membrane time constant, and τ_{ref} is the refractory period. Using

$$J_{\text{max}}^{(i)} = J(r_{\text{max}}^{(i)}), \quad (31)$$

where $J_{\text{max}}^{(i)}$ is the input current corresponding to the maximum firing rate of neuron i , the affine current model

$$J^{(i)}(\chi) = \kappa_i \langle e_i, \chi \rangle + \beta_i \quad (32)$$

was parameterized as

$$\kappa_i = \frac{J_{\text{max}}^{(i)} - 1}{1 - x_{\text{int}}^{(i)}}, \quad \beta_i = 1 - \kappa_i x_{\text{int}}^{(i)}. \quad (33)$$

The continuous membrane-potential dynamics of each LIF neuron in the populations (o_1, o_2, o_e) are described by

$$\tau_{\text{rc}} \frac{dV_i(t)}{dt} = -V_i(t) + J_i(t), \quad (34)$$

where $V_i(t)$ is the membrane potential of neuron i , and $J_i(t)$ is its input current. A spike is emitted when $V_i(t) \geq V_{\text{th}}$, where V_{th} is the firing threshold. After a spike, the membrane potential is reset as $V_i(t) \rightarrow V_{\text{reset}}$, where V_{reset} is the reset potential.

Spike trains were exponentially filtered to approximate postsynaptic currents:

$$\frac{d\alpha_i(t)}{dt} = -\frac{\alpha_i(t)}{\tau_{\text{syn}}} + \sum_q \delta(t - t_q), \quad (35)$$

where $\alpha_i(t)$ is the filtered spike activity of neuron i , τ_{syn} is the synaptic time constant, $\delta(\cdot)$ is the Dirac delta function, and t_q is the time of the q -th spike. The index q is used for spike times to avoid confusion with the gain parameters.

3.4. Experimental protocol and saved metrics

For each architecture and each number of sensory neurons, the experiment was repeated for 10 independent tests with different random initializations of neural encoders, intercepts, maximum firing rates, and trainable decoders.

For each run, eight summary characteristics were saved: simulation time, emitted spikes per simulated second, mean amplitude correlation, mean instantaneous-frequency correlation, mean phase-locking value, mean absolute phase difference, $\text{RMSE}(u_{\text{ref}}, o_1)$, and $\text{RMSE}(o_1, g)$, where RMSE denotes the root mean square error. The main simulation parameters are summarized in Table 1.

Table 1. Main simulation parameters used in the sensory-neuron sweep.

Parameter	Value
Number of independent tests	10
Sensory population size	$N_{\text{sens}} = 10, 60, \dots, 560$ (step 50)
Latent population size	$N_{\text{lat}} = 50$
Error population size	$N_{\text{err}} = 50$
Simulation time step	$\Delta t = 0.001$ s
Total simulation time	$T = 30$ s
Cue duration	$T_{\text{cue}} = 5$ s
Metric evaluation start time	$t_0 = T_{\text{cue}} + 0.5$ s
Synaptic time constant	$\tau_{\text{syn}} = 0.05$ s
Membrane time constant	$\tau_{\text{rc}} = 0.02$ s
Refractory period	$\tau_{\text{ref}} = 0.002$ s
Learning rate	$\eta = 10^{-3}$

The parameters in Table 1 fall into two groups. The neuronal time constants were fixed to biologically motivated values: the membrane time constant $\tau_{\text{rc}} = 0.02$ s and the refractory period $\tau_{\text{ref}} = 0.002$ s are typical of cortical pyramidal neurons, and the synaptic time constant $\tau_{\text{syn}} = 0.05$ s lies within the range of postsynaptic time constants commonly used in NEF models of cortical dynamics [4,12]. The remaining parameters – the learning rate η , the ridge regularization coefficients λ_{rec} and λ_e , the cue duration T_{cue} , and the total simulation time T – were selected empirically to ensure stable convergence of the reconstruction decoder while keeping the autonomous sensory dynamics bounded. The robustness of these choices is examined systematically in Section 4.3 by varying all principal neuronal, learning, regularization, and cue-related parameters.

3.5. Reconstruction-quality metrics

To quantitatively characterize the reconstruction of the autonomous sensory trajectory $u_{o_1}[n] \in \mathbb{R}^D$ by the top-down estimate $g[n] \in \mathbb{R}^D$, the agreement of amplitude, frequency, and phase characteristics was assessed for each coordinate d . For the oscillator, $d \in \{x, y\}$, whereas for the Lorenz attractor, $d \in \{x, y, z\}$. The observation interval was defined by $n\Delta t \geq t_0$, where

$$t_0 = T_{\text{cue}} + 0.5 \text{ s} \quad (36)$$

is the metric evaluation start time. Here, T_{cue} is the end of the cue-forcing interval, and the additional 0.5 s offset removes the initial transient after the cue is switched off.

To exclude the contribution of the constant component, signal centering was applied:

$$\begin{aligned} \tilde{u}_{o_1}[n] &= u_{o_1}[n] - \langle u_{o_1}[n] \rangle_n, \\ \tilde{g}[n] &= g[n] - \langle g[n] \rangle_n, \end{aligned} \quad (37)$$

where $\langle \cdot \rangle_n$ denotes averaging over the discrete observation interval under consideration.

Amplitude fidelity was estimated through the analytical representation of the signal introduced using the Hilbert transform $\mathcal{H}\{\cdot\}$ [23]:

$$\begin{aligned} L_{o_1}[n] &= \tilde{u}_{o_1}[n] + i\mathcal{H}\{\tilde{u}_{o_1}[n]\}, \\ L_g[n] &= \tilde{g}[n] + i\mathcal{H}\{\tilde{g}[n]\}. \end{aligned} \quad (38)$$

Here, $L_{o_1}[n]$ and $L_g[n]$ are the analytic signals of the autonomous sensory trajectory and of its top-down reconstruction, respectively, and i is the imaginary unit. The discrete construction of the analytic signal used in the numerical implementation is described in Appendix A.

The amplitude envelopes are defined as

$$A_{o_1}[n] = |L_{o_1}[n]|, \quad A_g[n] = |L_g[n]|. \quad (39)$$

The degree of amplitude matching was characterized by Pearson's linear correlation coefficient [24]:

$$\rho_A = \text{corr}(A_{o_1}, A_g). \quad (40)$$

Frequency fidelity was estimated using the instantaneous frequency $f[n]$ of the analytic signal. It was defined through the unwrapped phase angle $\phi[n]$ of the complex quantity $L[n]$:

$$f[n] = \frac{1}{2\pi} \frac{\phi[n] - \phi[n-1]}{\Delta t}. \quad (41)$$

Here, $L[n]$ denotes either $L_{o_1}[n]$ or $L_g[n]$, depending on the analyzed signal. To suppress high-frequency fluctuations, smoothing was applied using rectangular averaging over a window of length $N_{\text{win}} = 101$ samples, which corresponds to a time scale of approximately $N_{\text{win}}\Delta t \approx 0.101$ s. The matching of instantaneous frequencies was characterized by the correlation

$$\rho_f = \text{corr}(f_{o_1}, f_g), \quad (42)$$

where f_{o_1} and f_g are the instantaneous frequencies of u_{o_1} and g , respectively.

Phase fidelity was estimated through the phase difference

$$\Delta\phi[n] = \phi_{o_1}[n] - \phi_g[n], \quad (43)$$

where $\phi_{o_1}[n]$ and $\phi_g[n]$ are the unwrapped phases of $L_{o_1}[n]$ and $L_g[n]$, respectively. The phase-locking value (PLV) [25], which measures the stability of the phase difference between the sensory trajectory and its reconstruction, was then computed as

$$\text{PLV} = |\langle \exp(i\Delta\phi[n]) \rangle_{n \in \Omega}|, \quad (44)$$

taking values from 0 (the phase of the reconstruction is unrelated to that of the target) to 1 (the phase difference is constant, i.e. the phase is reproduced exactly up to a fixed offset). Additionally, an amplitude-confidence mask

$$\Omega = \{n : A_{o_1}[n] > \delta, A_g[n] > \delta\} \quad (45)$$

with a low threshold $\delta = 10^{-3}$ was introduced, limiting analysis to regions where the phase is determined stably. The average phase difference was determined as the phase of the complex average $\langle \exp(i\Delta\phi[n]) \rangle_{n \in \Omega}$. The value of the PLV above which the phase reconstruction was regarded as accurate was set to 0.95.

For multidimensional signals, all metrics were computed separately for each coordinate and then averaged over coordinates. In addition, two reconstruction errors were calculated:

$$\begin{aligned} \text{RMSE}(u_{\text{ref}}, o_1) &= \sqrt{\left\langle \|u_{\text{ref}}[n] - u_{o_1}[n]\|_2^2 \right\rangle_n}, \\ \text{RMSE}(o_1, g) &= \sqrt{\left\langle \|u_{o_1}[n] - g[n]\|_2^2 \right\rangle_n}. \end{aligned} \quad (46)$$

The first error characterizes the deviation of the autonomous sensory trajectory from the original reference signal, whereas the second error characterizes the accuracy of the top-down reconstruction.

4. Results

4.1. Oscillator reconstruction

The reconstruction quality for the oscillator was evaluated for the PC-SC and PC-EC architectures by varying the number of sensory neurons N_{sens} from 10 to 560. The results averaged over 10 independent tests are shown in Figure 2.

All reported simulations were performed under Windows 11 Pro 25H2 on a workstation equipped with an AMD Ryzen 7 7700 processor, 32 GB DDR5-4800 RAM, and NVMe solid-state storage. Simulations were executed entirely on the CPU and did not use GPU acceleration. The principal numerical software versions were NumPy 1.26.4, pandas 2.2.3, and psutil 7.1.3.

Both architectures achieve an accurate and stable reconstruction over the whole tested range. The amplitude correlation and the phase-locking value remain at ≈ 1.0 (the amplitude correlation is ≈ 0.95 at the smallest population, $N_{\text{sens}} = 10$, and ≥ 0.99 for every larger size), and the mean absolute phase difference stays below 3° for every population size, fluctuating between $\approx 2.2^\circ$ and $\approx 2.9^\circ$, which indicates that the top-down reconstruction g reproduces the temporal structure of the autonomous sensory dynamics o_1 . The RMSE between o_1 and g stays on the order of 10^{-2} throughout, remaining between ≈ 0.031 and ≈ 0.039 across the range, and confirms that the reconstruction decoder tracks the internally generated sensory representation with high accuracy. In all reconstruction-quality measures, the two architectures are nearly indistinguishable; their principal differences are the explicit spiking representation of the prediction error in PC-SC, the resulting increase in emitted spike rate, and the moderately longer simulation time. The instantaneous-frequency correlation is the single exception: it fluctuates irregularly between ≈ 0.75 and ≈ 1.0 , with a standard deviation reaching ≈ 0.35 , while every other measure is saturated. This is a property of the measure rather than of the model. For a strictly periodic reference, the instantaneous frequency of both o_1 and g is almost constant, so ρ_f is the linear correlation of two nearly constant sequences and is therefore dominated by residual numerical fluctuations rather than by any shared temporal structure. The measure is informative for the Lorenz attractor, where the instantaneous frequency genuinely varies over the trajectory, and is reported here only for completeness.

In contrast to the measures above, the RMSE between the reference trajectory and o_1 grows with the population size over the tested range, from ≈ 0.27 at $N_{\text{sens}} \approx 100$ to ≈ 0.70 at the largest population, $N_{\text{sens}} = 560$; the smallest population, $N_{\text{sens}} = 10$, gives a comparably large error (≈ 0.77), but for a different reason of insufficient representational capacity. This quantity does not measure how well g reconstructs o_1 ; it measures how far the autonomous sensory state has drifted away from the externally supplied reference. After the cue is removed, o_1 evolves under its own learned recurrent dynamics and is not

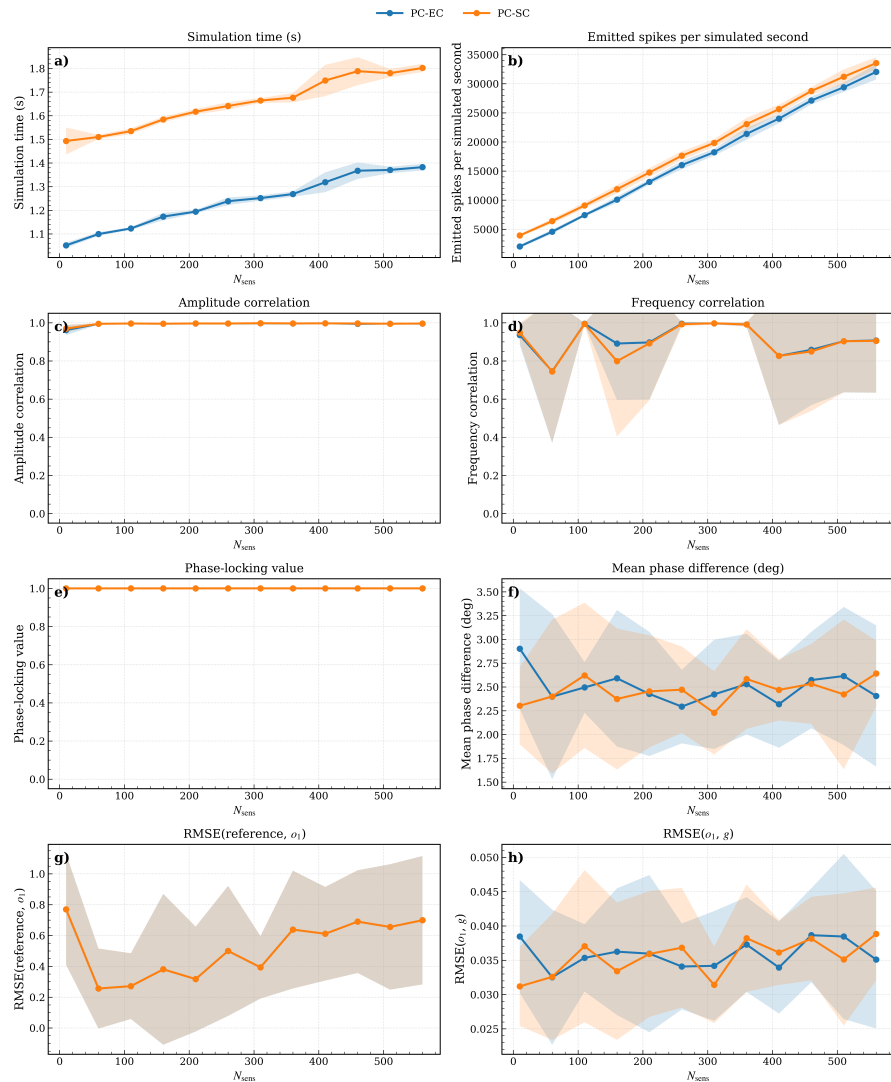


Figure 2. Comparison of PC-SC and PC-EC architectures for oscillator reconstruction as a function of the number of sensory neurons N_{sens} . (a) Simulation time; (b) emitted spikes per simulated second; (c) amplitude correlation; (d) frequency correlation; (e) phase-locking value (PLV); (f) mean phase difference; (g) RMSE between the reference trajectory and o_1 ; (h) RMSE between o_1 and g . Solid curves denote mean values and shaded regions denote standard deviations over 10 independent tests.

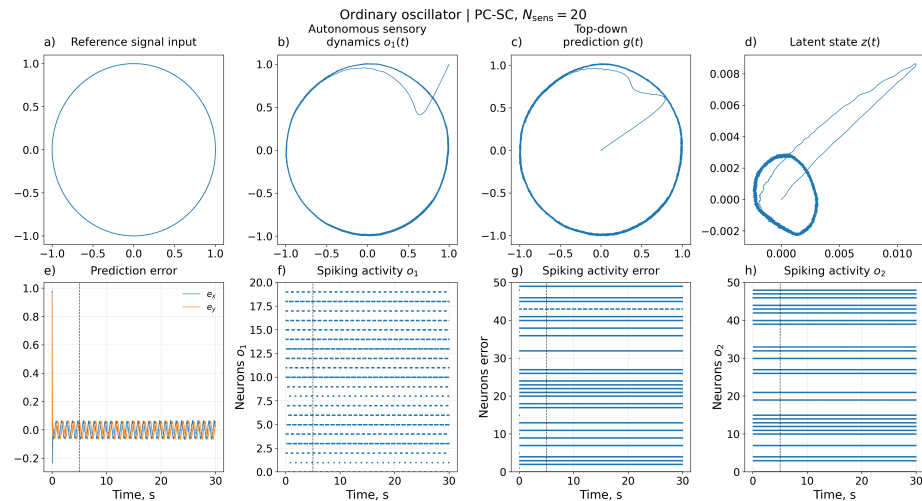


Figure 3. Qualitative analysis of oscillator reconstruction in the PC-SC architecture. (a) Reference oscillator trajectory; (b) autonomous sensory dynamics $o_1(t)$; (c) top-down reconstruction $g(t)$; (d) latent state $z(t)$; (e) prediction error e ; (f)–(h) raster plots of the sensory, error, and predictive populations. The vertical dashed line denotes the end of the cue period at $t = 5$ s.

forced to follow u_{ref} pointwise, so any mismatch in the reconstructed angular frequency accumulates into a phase drift.

Figure 3 illustrates the internal dynamics of the PC-SC architecture. After the cue period, the sensory population o_1 maintains stable autonomous oscillations, while the reconstruction g follows the same phase-space structure. The prediction error remains bounded after the initial transient, confirming that the predictive population reconstructs the sensory state accurately. The raster plots demonstrate that the sensory state, the prediction error, and the higher-level reconstruction are represented by separate spiking populations, which is the key feature of the PC-SC architecture.

4.2. Lorenz attractor reconstruction

The reconstruction quality for the Lorenz attractor was analyzed for the PC-SC and PC-EC architectures as a function of the number of sensory neurons N_{sens} . The results averaged over 10 independent tests are shown in Figure 4.

For the Lorenz attractor, the reconstruction-quality measures improve as the sensory population is increased and saturate by $N_{\text{sens}} \approx 150$ –200. At the smallest population tested ($N_{\text{sens}} = 10$) the reconstruction is only partial – the phase-locking value is ≈ 0.75 , the amplitude correlation ≈ 0.68 and the mean absolute phase difference $\approx 25^\circ$, but all three improve rapidly with N_{sens} : at $N_{\text{sens}} \approx 100$ the PLV is ≈ 0.95 , the amplitude correlation ≈ 0.97 and the phase difference $\approx 5^\circ$, and from $N_{\text{sens}} \approx 200$ onwards the PLV reaches ≈ 1.0 , the amplitude correlation ≈ 0.99 and the mean phase difference falls below $\approx 3^\circ$.

As for the oscillator, PC-SC and PC-EC are nearly indistinguishable in the reconstruction-quality measures. Their main differences are therefore not in reconstruction fidelity but in architecture and computational activity: PC-SC explicitly represents the prediction error using a dedicated spiking population and consequently exhibits a somewhat higher emitted spike rate and a longer simulation time. Thus, with respect to the fidelity with which the top-down estimate reconstructs the autonomous sensory dynamics, a population of roughly 150–200 sensory neurons is already sufficient, and further growth of N_{sens} produces little improvement in these measures.

The RMSE between o_1 and g remains on the order of 10^{-2} across the tested range, growing slowly from ≈ 0.009 at $N_{\text{sens}} \approx 100$ to ≈ 0.016 at $N_{\text{sens}} = 560$. The RMSE between

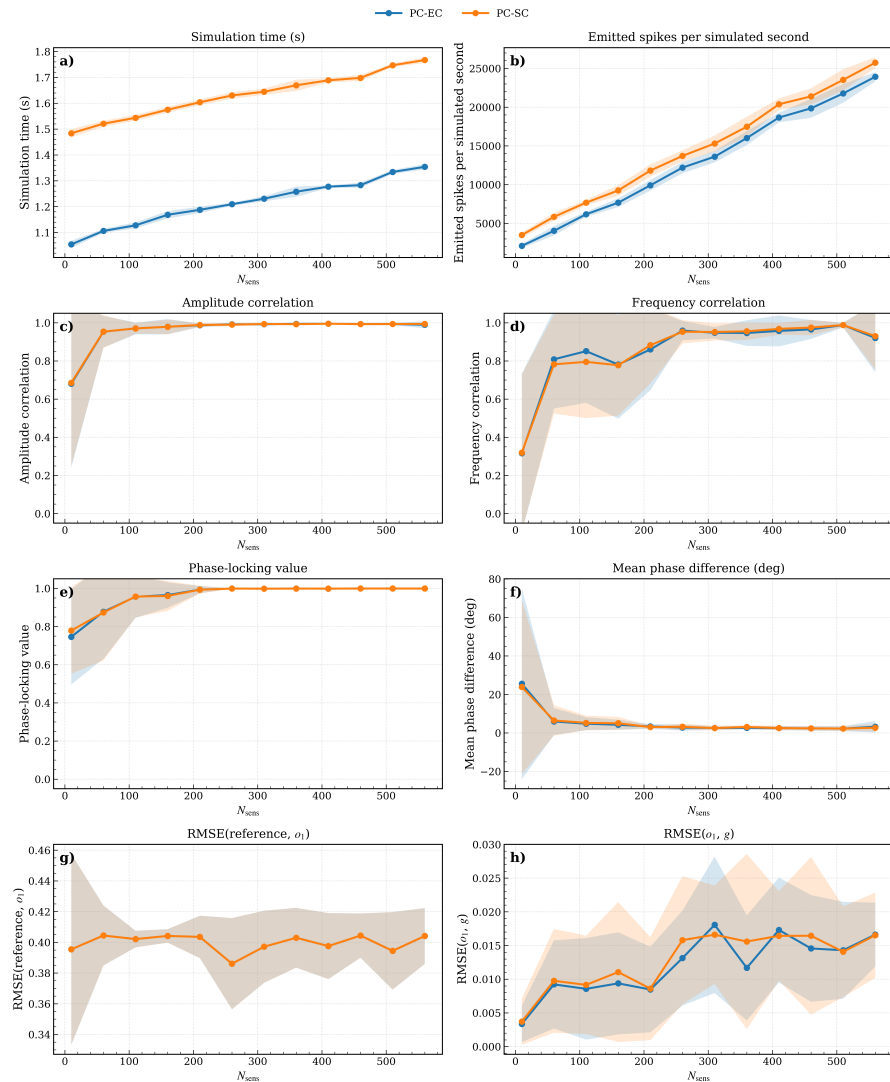


Figure 4. Comparison of PC-SC and PC-EC architectures for Lorenz attractor reconstruction as a function of the number of sensory neurons N_{sens} . (a) Simulation time; (b) emitted spikes per simulated second; (c) amplitude correlation; (d) frequency correlation; (e) phase-locking value; (f) mean phase difference; (g) RMSE between the reference trajectory and σ_1 ; (h) RMSE between σ_1 and g . Solid curves denote mean values and shaded regions denote standard deviations over 10 independent tests.

the reference trajectory and o_1 , by contrast, stays close to 0.40 (it fluctuates between ≈ 0.39 and ≈ 0.41) and is essentially independent of the population size. For a chaotic reference, any infinitesimal difference between o_1 and u_{ref} grows exponentially, so the two trajectories decorrelate within a few seconds of autonomous operation regardless of how faithfully the network reproduces the dynamics. Once they have decorrelated, the pointwise error is set by the geometry of the attractor alone: two independent trajectories on the same normalized Lorenz attractor give an RMSE of ≈ 0.38 – 0.49 under the same definition. The observed values coincide with this decorrelation ceiling, which means that $\text{RMSE}(u_{\text{ref}}, o_1)$ is saturated and cannot discriminate between a faithful and an unfaithful reconstruction of the chaotic dynamics.

The two architectures also differ in computational cost. For both benchmark systems, the simulation time increases moderately with the number of sensory neurons N_{sens} , while PC-SC remains consistently slower than PC-EC over the entire tested range. The additional run time of PC-SC is expected because this architecture contains an explicit spiking error population and therefore requires the simulation of an additional neural population and its associated transformations within the predictive-coding loop.

The emitted spike rate also increases approximately linearly with N_{sens} in both architectures. PC-SC consistently produces a somewhat higher overall spiking activity than PC-EC, which is consistent with the presence of the dedicated spiking error population. At the same time, the approximately linear growth of the spike rate with population size indicates that increasing representational capacity does not produce a disproportionate increase in network spiking activity over the tested range.

In terms of reconstruction quality, PC-EC and PC-SC perform comparably across the tested range of population sizes (Figures 2 and 4), and differ mainly in computational cost.

Figures 5 and 6 show that the autonomous sensory population preserves an attractor-like structure after the cue period, while g reconstructs the main geometry of this dynamics. The important result is that the predictive population o_2 continues to reconstruct the autonomous sensory dynamics: the low RMSE between o_1 and g , together with the high amplitude- and phase-fidelity measures, confirms that the PC-SC architecture achieves robust predictive reconstruction while preserving an explicit spiking representation of the prediction error.

It should also be noted that the dynamics of the latent state $z(t)$ do not coincide pointwise with the dynamics of the modeled signal. The latent space does not reproduce the full phase trajectory of the oscillator or Lorenz system directly; instead, it forms a compressed internal representation that captures the simplified structure required to drive the top-down reconstruction. Therefore, the role of the latent population is not to duplicate the sensory dynamics, but to provide a lower-dimensional predictive control signal from which the autonomous sensory representation $o_1(t)$ can be reconstructed as $g(t)$.

4.3. Hyperparameter sensitivity and stability analysis

We performed a one-at-a-time sensitivity analysis over eight principal hyperparameters of the model, sweeping each parameter while holding all others at their reported baseline values. The analysis was carried out for both reference systems – the Lorenz attractor ($D = 3$, $N_{\text{sens}} = 540$, Figure 8) and the 2-D oscillator ($D = 2$, $N_{\text{sens}} = 20$, Figure 7) – and for both predictive-coding architectures. The parameters considered were the PES learning rate η , the synaptic time constant τ_{syn} , the membrane time constant τ_{RC} , the refractory period τ_{ref} , the sensory recurrent-decoder regularization λ_{rec} , the error-decoder regularization λ_{e} , the cue duration T_{cue} , and the latent-state clipping bound c .

Every configuration was repeated over ten random seeds. The two architectures were evaluated using paired seeds so that differences between PC-EC and PC-SC were not

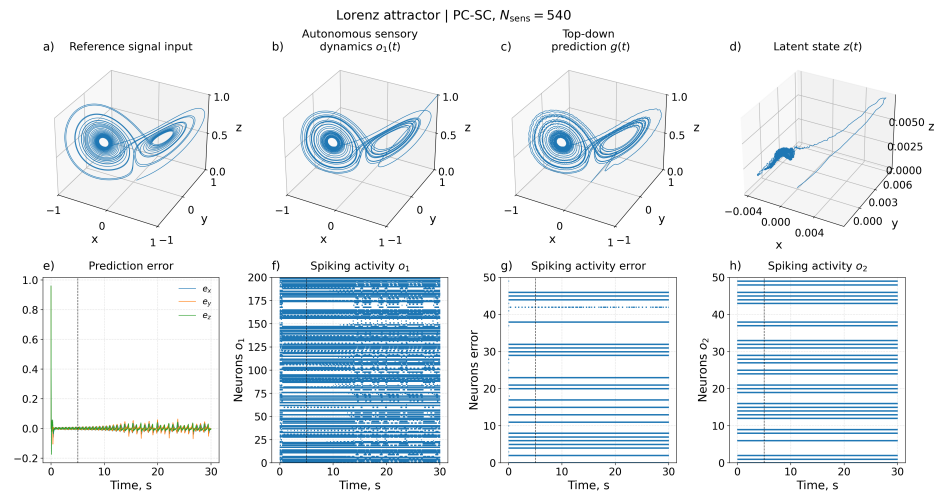


Figure 5. Qualitative analysis of Lorenz attractor reconstruction in the PC-SC architecture. (a) Normalized reference Lorenz trajectory; (b) autonomous sensory dynamics $o_1(t)$; (c) top-down reconstruction $g(t)$; (d) latent state $z(t)$; (e) prediction error e_t ; (f)–(h) raster plots of the sensory, error, and predictive populations. The vertical dashed line denotes the end of the cue period at $t = 5$ s.

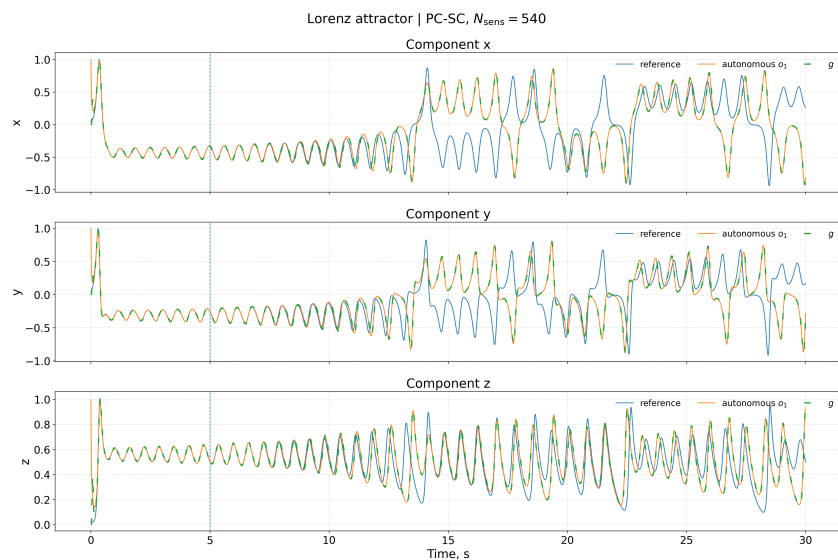


Figure 6. The component-wise temporal dynamics of Lorenz attractor reconstruction in the PC-SC architecture. The panels show the normalized x , y , and z components of the reference trajectory, autonomous sensory state $o_1(t)$, and reconstruction $g(t)$. After the cue period, $g(t)$ closely follows $o_1(t)$, demonstrating an accurate top-down reconstruction of the sensory population state, while both diverge from the reference as the chaotic trajectories decorrelate.

Table 2. One-at-a-time sensitivity analysis: change in $\text{RMSE}(o_1, g)$ across the tested range. “Baseline” is the value used elsewhere in the paper. $\text{RMSE}(u_{\text{ref}}, o_1)$ is discussed in the text and is omitted here.

Parameter	Sym.	Baseline	Range	Lorenz attractor	Oscillator
Learning rate	η	10^{-3}	$10^{-5} - 10^{-2}$	Flat to 10^{-3} ; PC-EC diverges at 10^{-2} , PC-SC unchanged.	Monotone fall 0.65 $\rightarrow$ 0.03; larger rates not tested.
Synaptic τ	τ_{syn}	0.05 s	0.01–0.20 s	Minimum at ≈ 0.02 s; $\times 3$ at 0.01 s.	Same minimum; total variation $\approx 25\%$.
Membrane τ	τ_{rc}	0.02 s	0.005–0.05 s	Monotone growth $\times 3.7 / \times 3.0$.	Monotone growth $\times 2.8$.
Refractory	τ_{ref}	0.002 s	0.001–0.004 s	Monotone fall $-45\% / -53\%$.	Monotone fall $-51\% / -46\%$.
Recurrent ridge	λ_{rec}	10^{-2}	$10^{-3} - 10^{-1}$	Monotone fall $\approx -30\%$.	Shallow maximum at the baseline; trades off against $\text{RMSE}(u_{\text{ref}}, o_1)$.
Error ridge	λ_e	10^{-2}	$10^{-3} - 10^{-1}$	None.	None.
Cue duration	T_{cue}	5 s	1–20 s	Rises $\approx +33\%$.	Flat.
Latent clipping bound	$\bar{c}$	1	0, 1, 2	No systematic degradation at $\bar{c} = 1$ relative to no clipping; differences remain within seed variability.	Reconstruction accuracy is essentially insensitive to the tested clipping bound.

confounded by different random population realizations. Markers in Figures 7 and 8 show the mean over seeds and error bars show one standard deviation. The dashed vertical line in each panel denotes the baseline value used in the main experiments.

Two reconstruction errors are reported. The left column shows $\text{RMSE}(u_{\text{ref}}, o_1)$ and therefore quantifies how closely the autonomous sensory population reproduces the original reference trajectory. The right column shows $\text{RMSE}(o_1, g)$ and directly measures the accuracy with which the top-down pathway reconstructs the current autonomous sensory dynamics. Because the dynamics of o_1 are generated independently of the PC-EC or PC-SC error pathway, the curves in the left column coincide for the two architectures. Consequently, $\text{RMSE}(o_1, g)$ is the more direct diagnostic of changes in the predictive-coding pathway, including changes in the latent-state projection.

For the Lorenz attractor, $\text{RMSE}(u_{\text{ref}}, o_1)$ remains of the same order, approximately 0.4, over most of the tested parameter ranges. This behaviour is expected for an autonomous chaotic trajectory: even a small difference in the reconstructed vector field produces exponential separation from the particular reference realization, so long-time pointwise RMSE becomes dominated by trajectory decorrelation rather than by the preservation of the attractor regime itself. In contrast, the oscillator shows a considerably stronger dependence of $\text{RMSE}(u_{\text{ref}}, o_1)$ on parameters such as τ_{syn} and T_{cue} , because small frequency errors accumulate as a systematic phase shift instead of producing chaotic decorrelation.

The PES learning rate has the strongest effect on reconstruction accuracy. At small learning rates the top-down decoder adapts too slowly, giving a large $\text{RMSE}(o_1, g)$. For the oscillator, the error decreases strongly as η is increased from 10^{-5} to the reported value $\eta = 10^{-3}$ (Figure 7(b)). The Lorenz system also remains well behaved around the reported operating point, whereas sufficiently large learning rates can destabilize the adaptive feedback loop, particularly for PC-EC (Figure 8(b)). The reported $\eta = 10^{-3}$ therefore

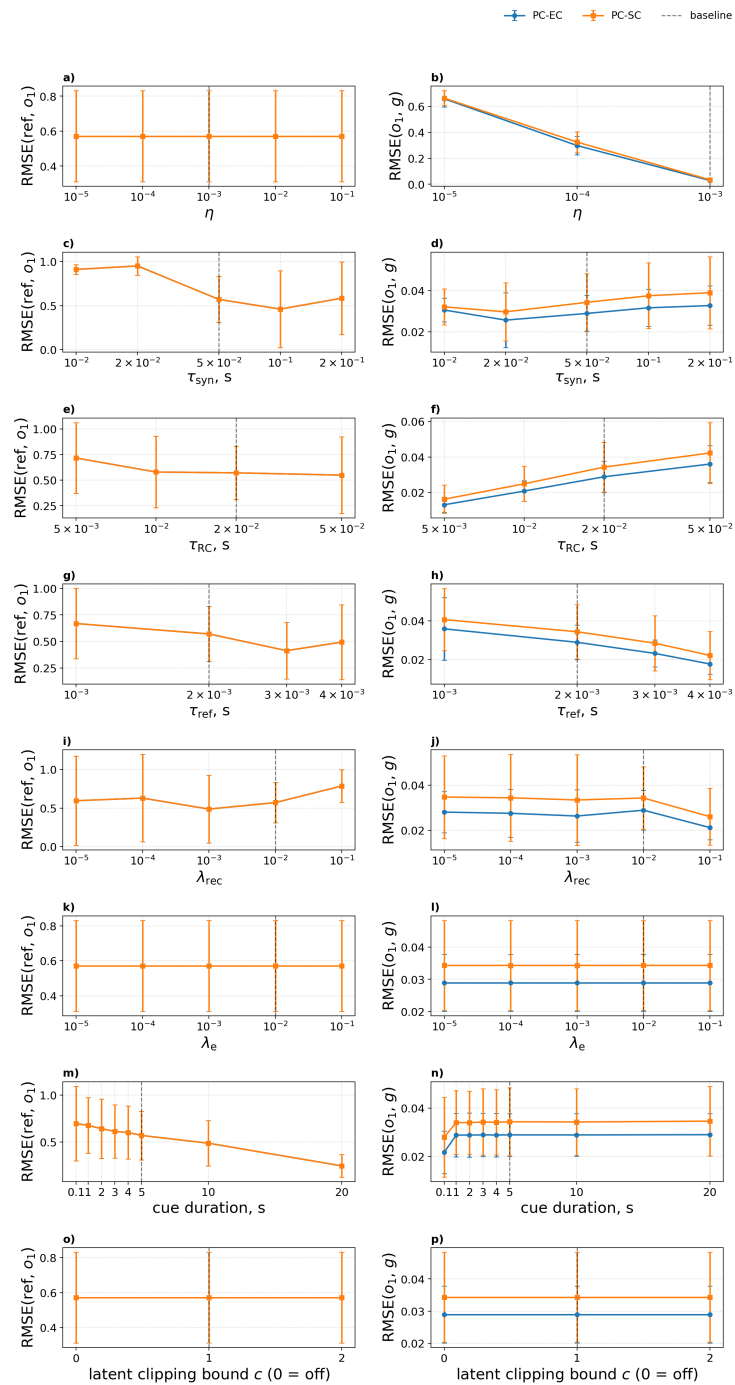


Figure 7. One-at-a-time hyperparameter sensitivity for the 2-D oscillator ($N_{\text{sens}} = 20$, ten seeds). Markers show means, error bars indicate standard deviations, and dashed vertical lines denote the reported baseline values. The left column shows $\text{RMSE}(u_{\text{ref}}, o_1)$ and the right column shows $\text{RMSE}(o_1, g)$. Rows correspond to η (a, b), τ_{syn} (c, d), τ_{RC} (e, f), τ_{ref} (g, h), λ_{rec} (i, j), λ_e (k, l), T_{cue} (m, n), and the latent-state clipping bound c (o, p), where $c = 0$ disables clipping. PC-EC is shown in blue and PC-SC in orange. In the left column, the two architectures coincide because the autonomous sensory dynamics are identical in the two models.

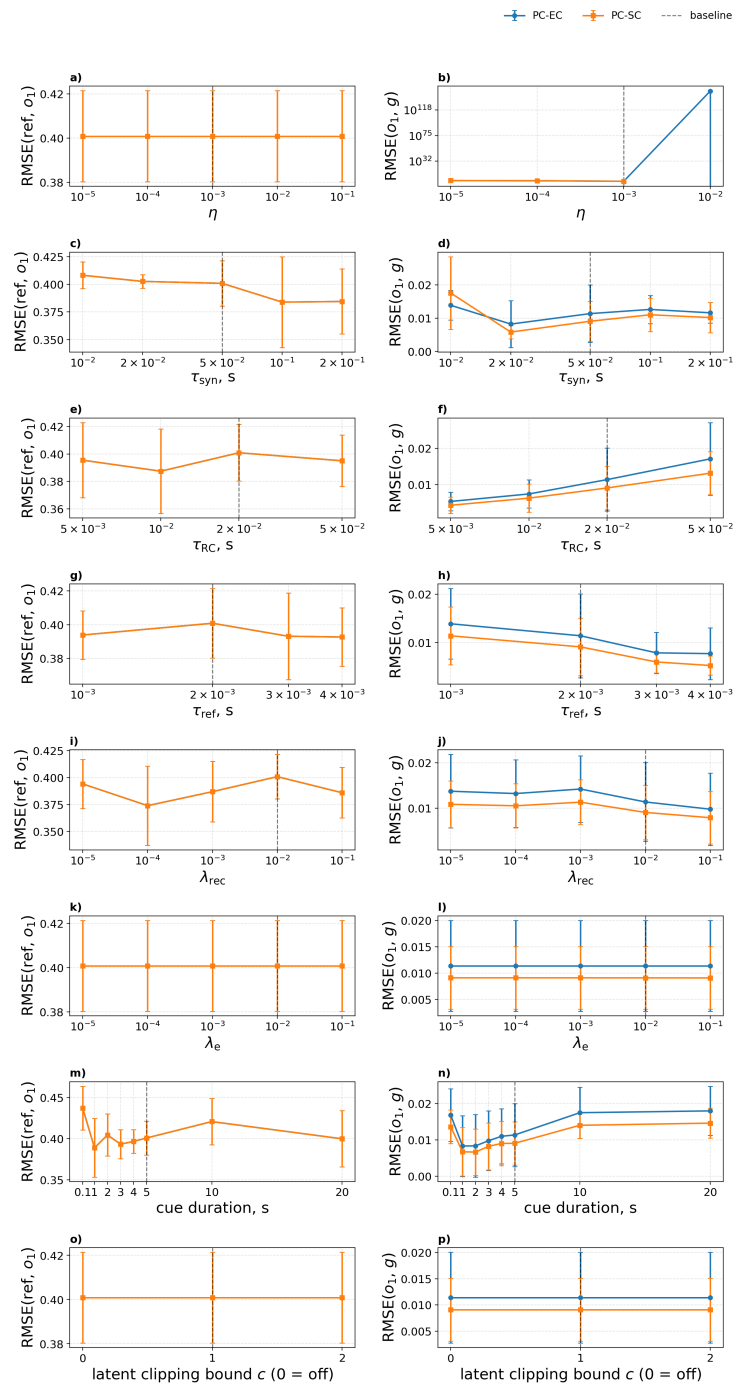


Figure 8. One-at-a-time hyperparameter sensitivity for the Lorenz attractor ($N_{\text{sens}} = 540$, ten seeds). Panel ordering, colors, error bars, baseline indicators, and error definitions are the same as in Figure 7.

provides a common stable operating point for both architectures rather than being selected solely as the minimum of a single error curve.

The difference between the architectures at large learning rates is consistent with the additional bounded representation introduced by the spiking error population in PC-SC. In PC-SC, the error is represented by the spiking population o_e ; thus, extreme prediction errors are filtered through a finite neuronal representation before entering the latent dynamics. Importantly, the same continuous prediction error is used by the PES decoder update in both architectures, so the observed difference arises from the error pathway rather than from a different learning rule.

The remaining neuronal time constants produce consistent trends in the two reference systems. Increasing τ_{RC} increases $\text{RMSE}(o_1, g)$, indicating that slower membrane integration reduces the temporal precision of the decoded top-down estimate. Over the tested range, increasing τ_{ref} decreases the reconstruction error, consistent with a reduction in high-frequency spike variability. Values substantially above the tested interval were not used because the maximum achievable firing rate of an LIF neuron is limited by $1/\tau_{ref}$ and would become incompatible with the firing-rate range used to construct the populations.

The synaptic time constant shows a non-monotonic effect. Very short filtering provides insufficient temporal averaging of the spike trains, whereas excessively slow filtering reduces the ability of the network to follow the reference dynamics. The resulting optimum therefore lies at a short-to-intermediate τ_{syn} . The oscillator additionally shows a pronounced dependence of $\text{RMSE}(u_{ref}, o_1)$ on τ_{syn} , because changes in the reconstructed autonomous frequency accumulate into phase error over time.

Regularization of the sensory recurrent decoder, λ_{rec} , has a smaller effect on $\text{RMSE}(o_1, g)$ than the neuronal time constants or the PES learning rate. Its influence is more apparent in the autonomous reconstruction error, particularly for the oscillator, reflecting the usual trade-off between a smoother recurrent solution and precise reproduction of the target dynamics. In contrast, varying λ_e produces almost no change in PC-SC over the tested interval. PC-EC contains no spiking error decoder and is therefore unaffected by this parameter by construction.

The cue-duration sweep confirms that the reported results are not tied to a narrow initialization interval. Changing T_{cue} modifies the initial condition from which the autonomous sensory trajectory is released, and therefore has a stronger effect on $\text{RMSE}(u_{ref}, o_1)$ than on the subsequent synchronization between o_1 and g . In particular, this distinction is important for the Lorenz system because two autonomous chaotic trajectories can rapidly decorrelate pointwise even while remaining on the same dynamical attractor.

Finally, we explicitly tested the influence of the bounded latent-state projection in order to determine whether clipping itself distorts the reconstructed dynamics. The projection was varied between no clipping ($c = 0$), the reported bound $z \in [-1, 1]$ ($c = 1$), and a wider bound $z \in [-2, 2]$ ($c = 2$), while all remaining model parameters were kept at their reported values. The error-to-latent coupling was likewise kept at its baseline value throughout this one-at-a-time analysis. For the Lorenz attractor, removing the projection or widening its interval does not produce a systematic reduction of $\text{RMSE}(o_1, g)$ relative to the reported $c = 1$ condition (Figure 8(p)). The differences between $c = 0$, $c = 1$, and $c = 2$ are small compared with the seed-to-seed variability. Thus, although the projection formally limits the latent representational domain, it does not measurably distort the reconstruction of the chaotic sensory dynamics at the reported operating point. The same result is observed for the 2-D oscillator (Figure 7(p)), where the three clipping conditions give essentially unchanged reconstruction errors.

The invariance of $\text{RMSE}(u_{\text{ref}}, o_1)$ to the clipping condition in Figures 8(o) and 7(o) is expected: the autonomous sensory population o_1 is generated upstream of the predictive latent state and is therefore not modified by the projection applied to z . The relevant test of possible clipping-induced distortion is instead $\text{RMSE}(o_1, g)$ in panels (p), for which no systematic loss of accuracy is observed.

These results do not imply that clipping is irrelevant for arbitrarily large latent drives. Rather, they show that at the reported operating point the projection acts as a representational safeguard without introducing a detectable reconstruction penalty. A substantially amplified error-to-latent gain would constitute a separate stress-test regime in which the projection could become active much more frequently; such a regime is distinct from the one-at-a-time sensitivity analysis reported here.

Overall, the reported parameter set is not selected by independently minimizing every reconstruction-error curve. In particular, smaller τ_{RC} or larger τ_{ref} can reduce $\text{RMSE}(o_1, g)$ within the tested ranges. The baseline values were chosen as a common biologically plausible operating point (Sec. 3.4). The sensitivity analysis shows that the main reconstruction result is preserved over broad parameter ranges, that the principal instability is associated with excessively aggressive PES learning, and, importantly, that the bounded latent-state projection does not introduce a measurable loss of reconstruction accuracy for either the periodic or the chaotic reference dynamics.

4.4. Reconstruction after freezing synaptic adaptation

To determine whether accurate reconstruction depends on continuous adaptation of the prediction decoder, we performed an additional experiment in which PES learning was disabled after an initial training interval. In the baseline configuration, the top-down decoder is continuously updated using the instantaneous prediction error. Therefore, close tracking between the autonomous sensory state $o_1(t)$ and the top-down prediction $g(t)$ could, in principle, be explained by continuous adaptation of the decoder weights rather than by maintenance of the previously learned mapping.

To test this possibility, the simulation was divided into three temporal regimes. During the initial interval, from 0 to 5 s, the reference signal is applied as a cue to initialize and stabilize the sensory population. At $t = 5$ s, the cue is removed and the sensory population $o_1(t)$ continues to evolve through its recurrent dynamics. PES adaptation remains active during the following interval, from 5 to 15 s. At $t = 15$ s, the PES learning rate is set to zero and the top-down decoder weights are completely frozen for the remaining 15 s of the simulation. Thus, during the 15–30 s interval, no further modification of the prediction decoder takes place.

Figure 9 illustrates this experiment for the Lorenz attractor using the PC-SC architecture with $N_{\text{sens}} = 540$. The black dashed vertical line indicates removal of the external cue at $t = 5$ s, while the red vertical marker indicates termination of PES learning at $t = 15$ s. After cue removal, the sensory population continues to generate the Lorenz-like dynamics without direct reference input. After PES is switched off, the neural populations remain dynamically active, but the top-down decoder is no longer modified.

The prediction error increases after PES adaptation is terminated, showing that continued online learning contributes to the accuracy of moment-to-moment reconstruction. Nevertheless, the top-down prediction $g(t)$ does not collapse when the decoder weights are frozen. Instead, it continues to exhibit a structured trajectory that retains the characteristic two-lobe organization of the Lorenz attractor. Therefore, reconstruction during the final interval cannot be attributed to continuous adjustment of the decoder weights at every time step.

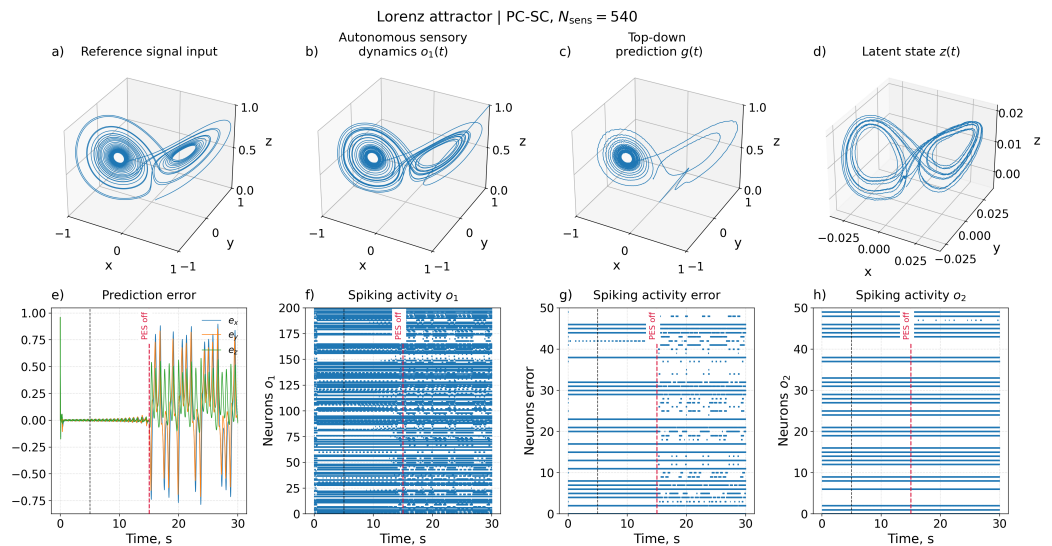


Figure 9. Frozen-decoder experiment for the Lorenz attractor using the PC-SC architecture ($N_{\text{sens}} = 540$). (a) Reference Lorenz trajectory. (b) Autonomous recurrent sensory dynamics $o_1(t)$. (c) Top-down prediction $g(t)$. (d) Latent state $z(t)$. (e) Prediction error. (f) Spiking activity of the sensory population o_1 . (g) Spiking activity of the prediction-error population. (h) Spiking activity of the latent/top-down population o_2 . The black dashed line at $t = 5$ s denotes removal of the external cue, whereas the red vertical marker at $t = 15$ s denotes termination of PES adaptation. During the subsequent 15–30 s interval, the decoder weights remain fixed, allowing reconstruction under active online learning to be distinguished from reconstruction using frozen learned weights.

Comparison of these two intervals makes it possible to distinguish the contribution of ongoing adaptation from the predictive behavior retained after learning has stopped. An increase in reconstruction error after $t = 15$ s is expected because the decoder can no longer compensate for instantaneous deviations through weight updates. At the same time, persistence of a structured prediction after freezing demonstrates that the reported reconstruction is not solely the result of continuous online regression.

It should be noted that switching off PES establishes autonomy with respect to decoder adaptation, but does not completely remove the prediction-error feedback pathway. In the PC-SC architecture, the instantaneous prediction error continues to be represented by the spiking error population and continues to influence the latent state $z(t)$ after the decoder weights have been frozen. Consequently, this experiment should be interpreted as a frozen-decoder predictive reconstruction within a closed error-correcting loop, rather than as completely open-loop generation by an isolated internal model. Nevertheless, the experiment directly addresses the concern that the observed reconstruction could be explained solely by a fast adaptive filter, because the prediction persists for an extended interval in which no further decoder learning occurs.

4.5. PC-EC versus PC-SC

The studied architectures differ in the size of the region in which they work, not in the accuracy they achieve inside it. With respect to accuracy, they are equivalent for practical purposes, and the residual differences do not point the same way: PC-SC lies 15–30% below PC-EC at every point of the Lorenz system sweeps and 15–20% above it at every point of the oscillator sweeps. Both differences are smaller than the seed-to-seed spread, and because their sign depends on the reference system, neither should be presented as an architectural property.

The stability is not symmetric. Seven of the eight parameters leave both models in the regime of accurate predictive reconstruction over their entire ranges in both systems; the seventh parameter removes PC-EC from that regime while leaving PC-SC untouched. The contribution of the explicit spiking error representation is therefore a larger stable operating region for the predictive-coding loop rather than a more accurate reconstruction, obtained at a simulation time $\approx 30\text{--}40\%$ higher than PC-EC's (Sec. 4).

5. Conclusion

Our results show that predictive-coding mechanisms implemented within the NEF framework can reconstruct the state of one neural population from the activity of another for both regular oscillations and Lorenz attractor chaotic dynamics. Numerical analysis demonstrates that increasing the population size improves the agreement between the reconstructed signal and the target trajectory: amplitude correlation increases, instantaneous-frequency agreement improves, and phase fidelity reaches near-limiting values when a sufficient number of neurons is used.

The comparison of the two architectures shows that both the error-driven control architecture and the spiking predictive coding architecture are capable of stable predictive reconstruction. The error-driven control model is computationally simpler and faster, since the prediction error is represented as a continuous control signal. In contrast, the spiking predictive coding architecture requires longer simulation time, but provides an explicit spiking population-level representation of the prediction error in the bottom-up state-update pathway.

It should also be emphasized that the latent state does not reproduce the modeled signal pointwise. Its dynamics may differ from the phase trajectory of the oscillator or Lorenz attractor, because the latent space acts as an internal predictive-control representation rather than as a direct copy of the sensory dynamics. Thus, the latent population encodes a simplified structure that is sufficient to drive the top-down prediction and to maintain an accurate reconstruction of the autonomous sensory state. The dynamics of the latent space can serve as an input signal for the next hierarchical level.

The predictive coding combined with PES learning provides a viable framework for dynamic control and state reconstruction in spiking neural networks. The proposed approach can be useful for neurobiological modeling, analysis of predictive-reconstruction mechanisms in neural populations, and the development of neuromorphic systems.

Future work will focus on extending the present reconstruction architecture toward a more complete spiking implementation of predictive coding. Also, proper evaluation of sensory noise, spike loss and temporary input occlusion are necessary on higher-dimensional nonlinear systems and real temporal data.

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Abbreviations

The following abbreviations are used in this manuscript:

LIF	Leaky Integrate-and-Fire
NEF	Neural Engineering Framework
PC	Predictive coding
PC-EC	Predictive-coding architecture with an external comparator
PC-SC	Predictive-coding architecture with a spiking comparator
PES	Prescribed Error Sensitivity
PLV	Phase Locking Value
RMSE	Root Mean Square Error
SNN	Spiking neural network
STDp	Spike-timing-dependent plasticity

Appendix A. Discrete construction of the analytic signal

This appendix describes the discrete implementation of the analytic signal used in the reconstruction-quality metrics. The procedure is used to compute the amplitude envelope, instantaneous phase, instantaneous frequency, and phase-locking value from the simulated time series. In the main text, these quantities are used to compare the autonomous sensory trajectory $u_{o_1}[n]$ with its top-down reconstruction $g[n]$.

In discrete form, a Fourier transform is first performed:

$$X[f] = \sum_{t=0}^{N-1} x[t] e^{-i2\pi ft/N}, \quad f = 0, \dots, N-1. \quad (\text{A1})$$

Negative frequencies are set to zero, positive frequencies are doubled, and $f = 0$ and, for even N , the Nyquist frequency $f = N/2$, remain undoubled. For this purpose, a frequency multiplier $H_N[f]$ is introduced:

$$H_N[f] = \begin{cases} 1, & f = 0, \\ 2, & 1 \leq f \leq \lfloor \frac{N-1}{2} \rfloor, \\ 1, & f = \frac{N}{2} \text{ and } N \text{ is even,} \\ 0, & \text{otherwise.} \end{cases} \quad (\text{A2})$$

To calculate the analytic signal, an inverse Fourier transform is performed:

$$L[t] = \text{IFFT}\{X[f] \cdot H_N[f]\} = \frac{1}{N} \sum_{f=0}^{N-1} X[f] H_N[f] e^{i2\pi ft/N}. \quad (\text{A3})$$

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